

Dilution-dependent antibacterial mechanisms of honey: a systematic review and cross-study quantitative evidence synthesis

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Abstract

Honey exhibits broad-spectrum antibacterial activity, but whether its mechanism of action changes with dilution — rather than simply weakening — remains incompletely resolved. We conducted a systematic review and cross-study quantitative evidence synthesis. Searches of PubMed, Europe PMC and Crossref, plus citation mining and data-repository searches, retrieved 4,042 records; 2,237 unique records were screened, 412 studies met eligibility criteria, and structured data were extracted from 103 open-access full texts (4,650 observations). All 10 extractable antibacterial concentration series increased monotonically with honey concentration (median Spearman $\rho = 1.00$), while solution pH rose on dilution ($\rho = -1.00$), consistent with weakening acidity. Natural honeys exceeded sugar-equivalent artificial controls in 16 of 21 matched pairs (median potency ratio 1.38, $p = 0.005$). Catalase-treated honey retained a median 92% of activity (13 pairs, 2 studies). Cross-study associations between honey chemistry and outcomes were detectable within studies but did not generalize: leave-one-study-out cross-validated R^2 was negative for chemistry-based and label-based models alike. Directly matched dilution-resolved hydrogen peroxide and antibacterial measurements were rare. The evidence is consistent with dilution-dependent reweighting of osmotic, acidic, peroxide and non-peroxide mechanisms, but does not establish a universal mechanistic switch; dilution-resolved mechanistic measurement remains the key evidence gap.

Keywords: honey; antibacterial activity; dilution; hydrogen peroxide; methylglyoxal; water activity; systematic review

1. Introduction

Honey is a complex food matrix with well-documented in vitro antibacterial activity (Kwakman et al., 2010; Willix et al., 1992). Its activity arises from several partially independent mechanisms: high osmolarity and low water activity constrain microbial growth, its acidity (typically pH 3.2–4.5) inhibits many pathogens, bee-derived glucose oxidase generates hydrogen peroxide (H_2O_2) under aqueous conditions, and non-peroxide constituents — most prominently methylglyoxal (MGO) in manuka-type honeys — contribute additional activity (Kwakman et al., 2010; Kwakman et al., 2011).

These mechanisms respond differently to dilution. Osmotic and acidic constraints weaken progressively as honey is diluted, whereas glucose oxidase is inactive in undiluted honey and produces H₂O₂ only after partial dilution, as shown experimentally by Bang and colleagues (Bang et al., 2003). Non-peroxide components have their own concentration-response behaviour. It is therefore plausible that the dominant antibacterial mechanism of honey is dilution-dependent: the mechanism that explains activity at 80% honey may not be the mechanism that explains activity at 10% honey.

This question is relevant to food microbiology and food functionality, where honey is evaluated as a natural antimicrobial ingredient or surface treatment and is frequently used in diluted form. If high-concentration activity rests largely on constraints that collapse on dilution while peroxide-mediated activity appears only at intermediate dilutions, then assays performed on concentrated honey could systematically misestimate performance under realistic, diluted conditions. Because individual studies rarely measure all candidate mechanisms across a dilution gradient, cross-study synthesis is needed to assess the pattern.

We conducted a systematic review and quantitative evidence synthesis of primary studies addressing the dilution dependence of honey's antibacterial mechanisms. We asked: (i) how antibacterial activity varies with honey concentration; (ii) whether natural honey exceeds sugar-equivalent artificial-honey controls; (iii) how H₂O₂ production varies with dilution and whether it tracks measured activity; (iv) how much activity survives catalase-mediated H₂O₂ removal; (v) whether measured physicochemical parameters predict antibacterial outcomes beyond honey-type labels; and (vi) how results vary across organisms. We pre-specified that causal or mechanistic language would be reserved for within-study perturbation evidence (for example catalase treatment or artificial-honey comparators), with cross-study associations reported as observational.

2. Material and methods

2.1. Protocol and reporting

The review was conducted and reported with reference to the PRISMA 2020 guideline where applicable; the study flow is summarized in Fig. 1. The review was not prospectively registered; eligibility criteria, analysis decisions and the complete executable pipeline are released with this article (see Data availability).

2.2. Search strategy

We searched PubMed (NCBI E-utilities), Europe PMC (REST API) and Crossref (works API) for records combining honey-related terms with antibacterial, minimum inhibitory concentration (MIC), zone-of-inhibition, peroxide, methylglyoxal, water-activity and dilution terms; the full query strings, execution dates and hit counts are reported in Supplementary Table S1. Searches were executed programmatically and raw responses retained. Backward citation mining through Europe PMC contributed additional records, and Dryad, Figshare and Zenodo were searched for associated research data. All records were deduplicated on DOI, PubMed ID and normalized title.

2.3. Eligibility criteria

Records were eligible if they reported a primary in vitro antibacterial evaluation of natural honey with a quantitative endpoint (MIC, minimum bactericidal concentration (MBC), inhibition zone, biofilm inhibition or equivalent) and at least one of: a concentration or dilution series; an artificial-honey or sugar-equivalent comparator; a catalase or other mechanistic perturbation; or quantitative physicochemical characterization (H₂O₂, MGO, water activity, pH, phenolics). Reviews, non-antibacterial studies, non-honey studies, formulation studies in which honey could not be separated from the matrix, and records without a usable abstract were excluded. Eligibility was assigned by a deterministic rule-based classifier applied to titles and abstracts, followed by manual verification of included candidates; the classifier rules and per-record decisions are released with the dataset.

2.4. Data extraction

Open-access full texts were retrieved from PubMed Central as JATS XML. All tables were parsed programmatically, including nested tables, with merged header cells expanded and multi-row headers reconstructed. A rule-based extractor converted the dominant table layouts into long-format observations carrying study, table, row and column provenance. For key mechanistic datasets (catalase-treated versus untreated pairs, simulated-honey controls, concentration series and MGO-spike dose-response), rows were additionally curated manually after direct inspection of source tables and captions; manually curated rows are flagged and take precedence. Organism labels were harmonized to organism groups using a documented mapping. Concentrations embedded in sample labels were parsed into a dedicated field, and the reported concentration basis (percent weight/volume, volume/volume, weight/weight, mass concentration or unspecified) was preserved in a separate field; analyses were stratified by basis rather than pooled across incompatible units. Values reported with inequality signs were retained with a censoring flag.

2.5. Risk-of-bias and reporting-quality assessment

No validated risk-of-bias instrument exists for cross-study in vitro antibacterial data. We applied a structured framework covering five domains — assay standardization, comparator quality, dispersion reporting, mechanistic perturbation and reporting completeness — each scored 0-2. Availability of concentration-series data was recorded as an availability flag, not a quality judgement. Ratings (low, some or high concern) were derived from domain totals and used in sensitivity analyses.

2.6. Statistical analysis

Concentration-response was summarized by per-series Spearman correlations between concentration and response; a series was eligible when it contained at least four ordered concentrations with a monotone-reported endpoint. Natural versus artificial honey was compared within study, organism, outcome and concentration strata; to make inhibition-zone and MIC/MBC contrasts directionally comparable, effects were expressed as a potency ratio (natural/artificial for zones; artificial/natural for MIC/MBC, so that values above 1 always indicate greater natural-honey potency) and tested with a Wilcoxon signed-rank test on log ratios. Catalase-treated versus untreated observations were paired identically and expressed as residual activity fractions. Honey-level physicochemical parameters were joined to outcomes on study and

normalized sample label; Spearman correlations were computed per parameter and outcome, and ordinary least squares models on $\log_{10}(\text{MIC})$ compared measured chemistry, a honey-type label and their combination. Because rows from the same study are not independent, model performance was evaluated by leave-one-study-out cross-validation rather than in-sample fit. Organism heterogeneity was assessed with Kruskal-Wallis tests. Sensitivity analyses repeated key contrasts on the low-risk-of-bias subset, on each concentration basis separately, and after excluding manuka-labelled honeys; leave-one-study-out resampling was used to assess dependence on individual studies. Analyses were performed in Python 3.12 (pandas, SciPy, statsmodels); the released analysis scripts regenerate every reported number.

3. Results

3.1. Study selection and characteristics

Searches retrieved 4,042 records; after removing 1,805 duplicates, 2,237 records were screened and 1,372 excluded at title/abstract stage. Of 865 reports assessed, 412 met inclusion criteria (Fig. 1). Open-access full texts were obtained for 230 records, and quantitative data were extracted from 103 studies, yielding 4,650 harmonized observations (2,378 MIC, 1,362 inhibition-zone and 628 MBC values; the remainder comprise biofilm, phenol-equivalence and dilution-series measures). Physicochemical data were extracted for 59 studies. Risk-of-bias ratings were 'some concerns' for most extracted studies (93 of 103), reflecting incomplete dispersion reporting and inconsistent mechanistic characterization; 7 were rated low concern and 3 high concern.

The extracted subset is not a census of eligible studies. Of 412 eligible studies, 162 had no resolvable PubMed Central identifier, 20 had an identifier but no open-access full text, and 127 retrieved full texts yielded no machine-extractable outcome table meeting our parsing rules. Extracted studies were more recent than non-extracted studies (median year 2022 vs. 2020; Mann-Whitney $p < 0.001$) and overwhelmingly open-access (99% vs. 70% flagged open-access). Mechanistic content, however, was similar between the groups — dilution-related abstracts appeared in 34% of extracted versus 33% of non-extracted studies, peroxide-related in 21% versus 23%, and MGO-related in 17% versus 16% — indicating that selection operated mainly on access rather than on topic. Quantitative results below therefore describe the extractable subset; the selection mechanism is reported explicitly rather than treating the subset as a census of eligible work.

PRISMA 2020-style study flow

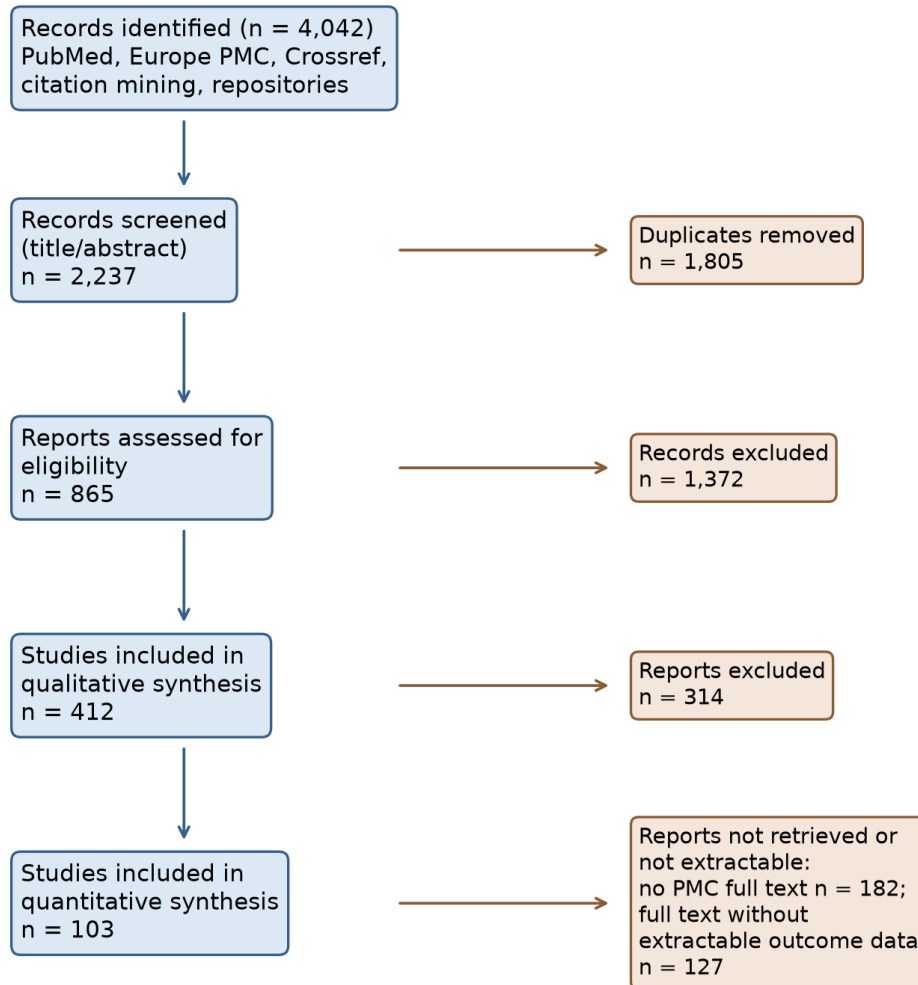


Fig. 1. PRISMA 2020-style flow of study identification, screening, eligibility and quantitative extraction.

3.2. Concentration-response

10 antibacterial concentration series were extractable across 3 studies, spanning inhibition zones, biofilm inhibition and carbapenemase-producer isolates. All 10 series increased monotonically with honey concentration (median Spearman $\rho = 1.00$; Fig. 2A-2C). In the one study reporting pH across a full dilution ladder, pH rose monotonically as honey was diluted for all four honeys examined ($\rho = -1.00$; Fig. 2D), consistent with progressive weakening of the acidic constraint. Fitting $\log(\text{response})$ on $\log(\text{concentration})$ across the zone series gave a slope of 0.33 (SE 0.55); the sub-linear point estimate is compatible with a saturating diffusion-assay response but the interval is wide and should not be over-interpreted. Concentration-series data remained scarce, so these results describe direction rather than a pooled dose-response curve.

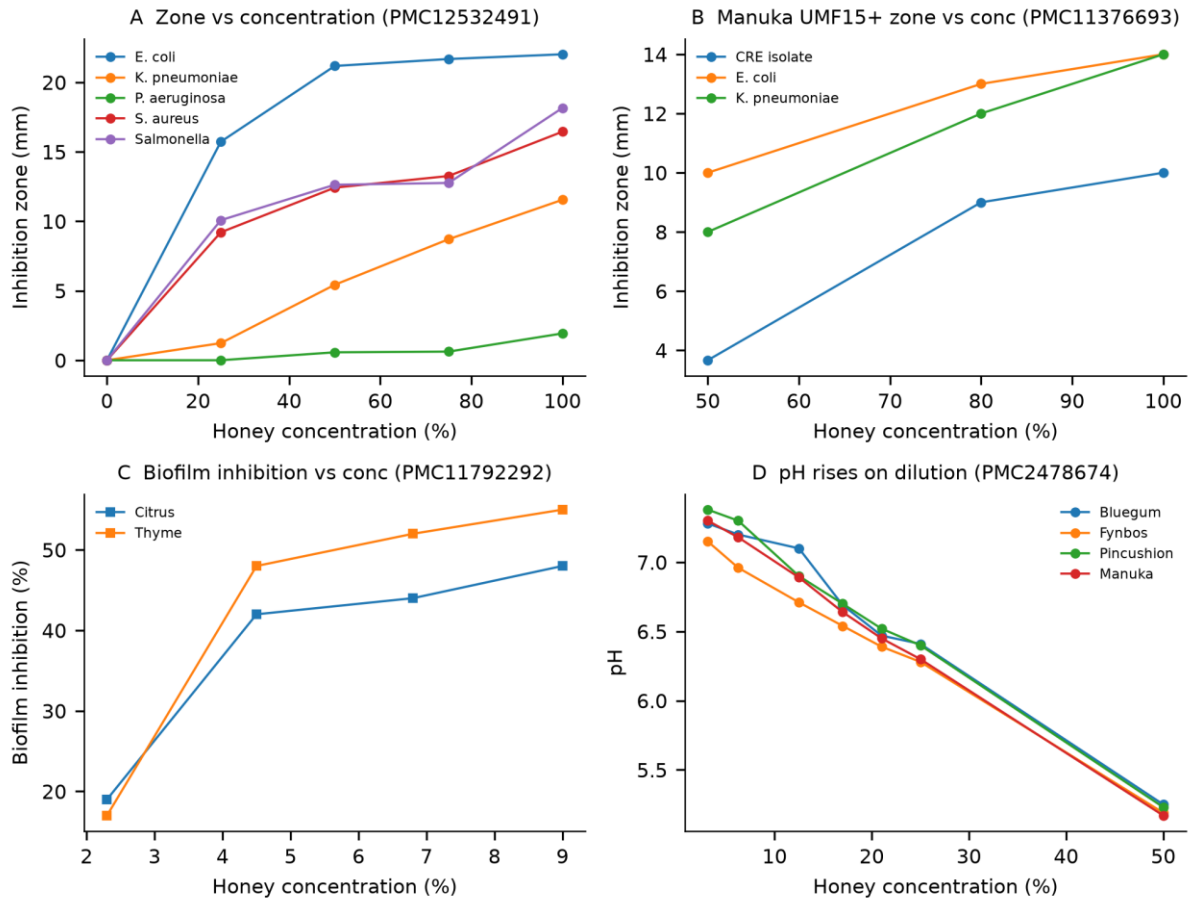


Fig. 2. Concentration-response series. (A) Inhibition zones versus honey concentration for five organisms (Kumera et al., 2025). (B) Manuka UMF15+ zones versus concentration for three organisms (Bukhari et al., 2024). (C) Biofilm inhibition versus concentration for two honeys (Alshaybawee et al., 2025). (D) pH versus concentration for four honeys (Basson and Grobler, 2008); pH rises as honey is diluted.

3.3. Natural versus artificial honey

21 paired comparisons of natural honey versus artificial or sugar-equivalent controls were identified (10 zone pairs, 11 MIC/MBC pairs). Direction-normalized, natural honey was more potent in 16 of 21 pairs (median potency ratio 1.38; Wilcoxon on log ratios, $p = 0.005$; Fig. 3). In zone units the median paired excess was +0.6 mm; in MIC units the median natural:artificial ratio was 0.81. Artificial honey thus reproduced part, but not all, of natural honey's effect — supporting a real osmotic contribution while indicating additional non-osmotic activity in most, though not every, study.

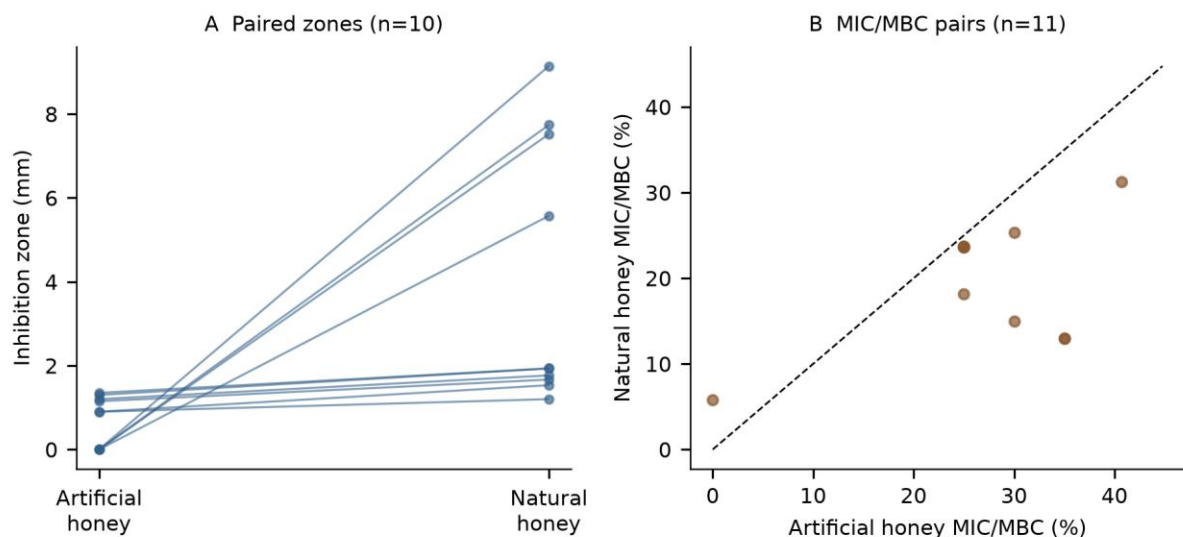


Fig. 3. Natural versus artificial honey. (A) Paired inhibition zones. (B) Paired MIC/MBC values; points below the diagonal indicate greater potency of natural honey.

3.4. Hydrogen peroxide and dilution

Directly matched dilution-resolved evidence on H₂O₂ was sparse. Within the extracted corpus, one study reported H₂O₂ detectable only at 6.25% and 3.12% honey and not at higher concentrations, and a second linked glucose oxidase activity, catalase sensitivity and H₂O₂ concentration to MIC; both are consistent with a peroxide mechanism that activates on dilution rather than at full strength. Across studies, honey-level H₂O₂ measurements correlated weakly and positively with MIC ($\rho = 0.46$, $n = 29$, 2 studies; Fig. 4) — a direction opposite to a naive 'more peroxide, more potent' expectation and most plausibly reflecting between-study confounding (potent non-peroxide honeys can be low in H₂O₂). We found no extracted dataset reporting a full H₂O₂ dilution ladder with matched antibacterial endpoints; the predicted intermediate-dilution peroxide peak is therefore supported only qualitatively, and we report it as an explicit evidence gap.

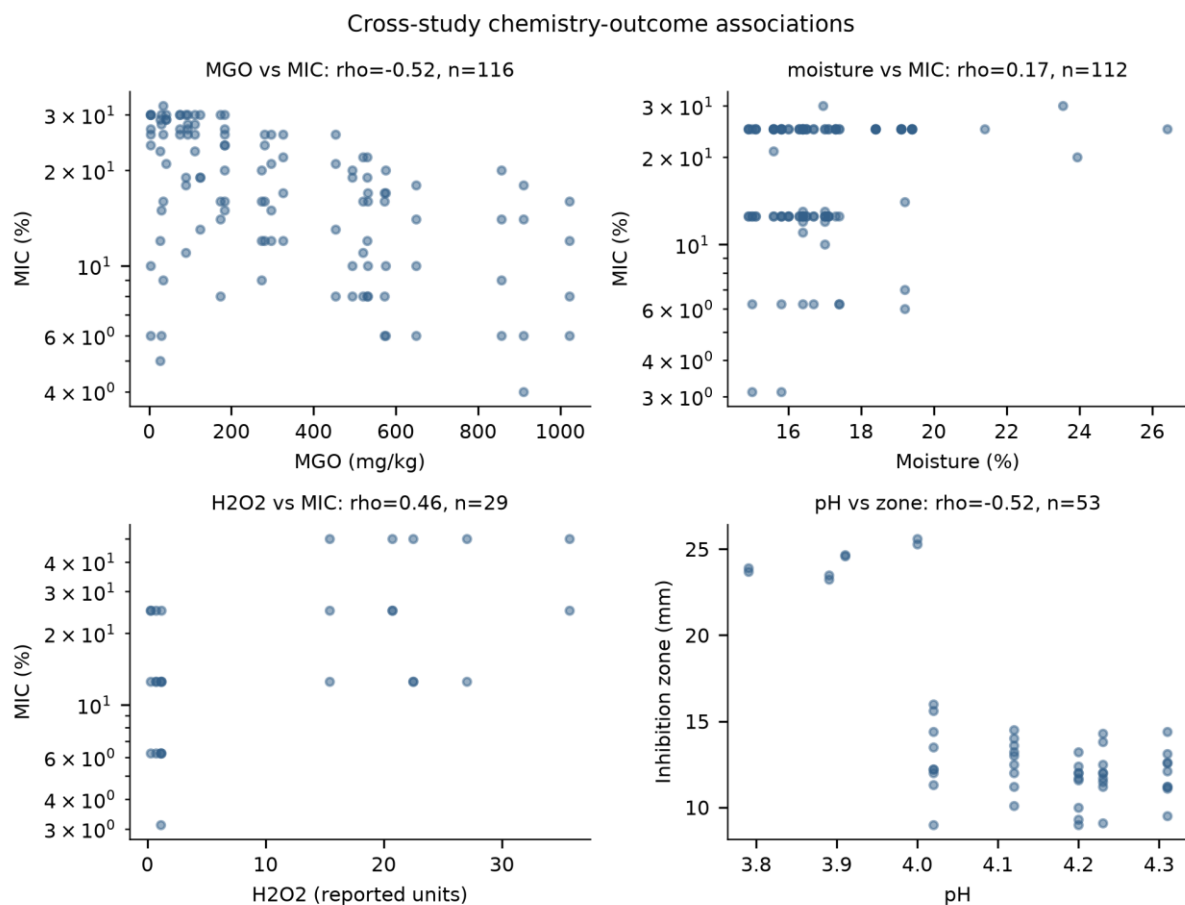


Fig. 4. Cross-study associations between honey-level physicochemical parameters and antibacterial outcomes (Spearman ρ ; MIC on a log scale).

3.5. Catalase perturbation and residual activity

13 paired catalase-perturbed observations were extracted from 2 studies. Catalase-treated honey retained a median 92% of untreated antibacterial activity (Fig. 5). This indicates substantial peroxide-independent activity in the honeys tested; with only two contributing studies the estimate is preliminary, and residual activity cannot be attributed to MGO or any other single constituent without further evidence.

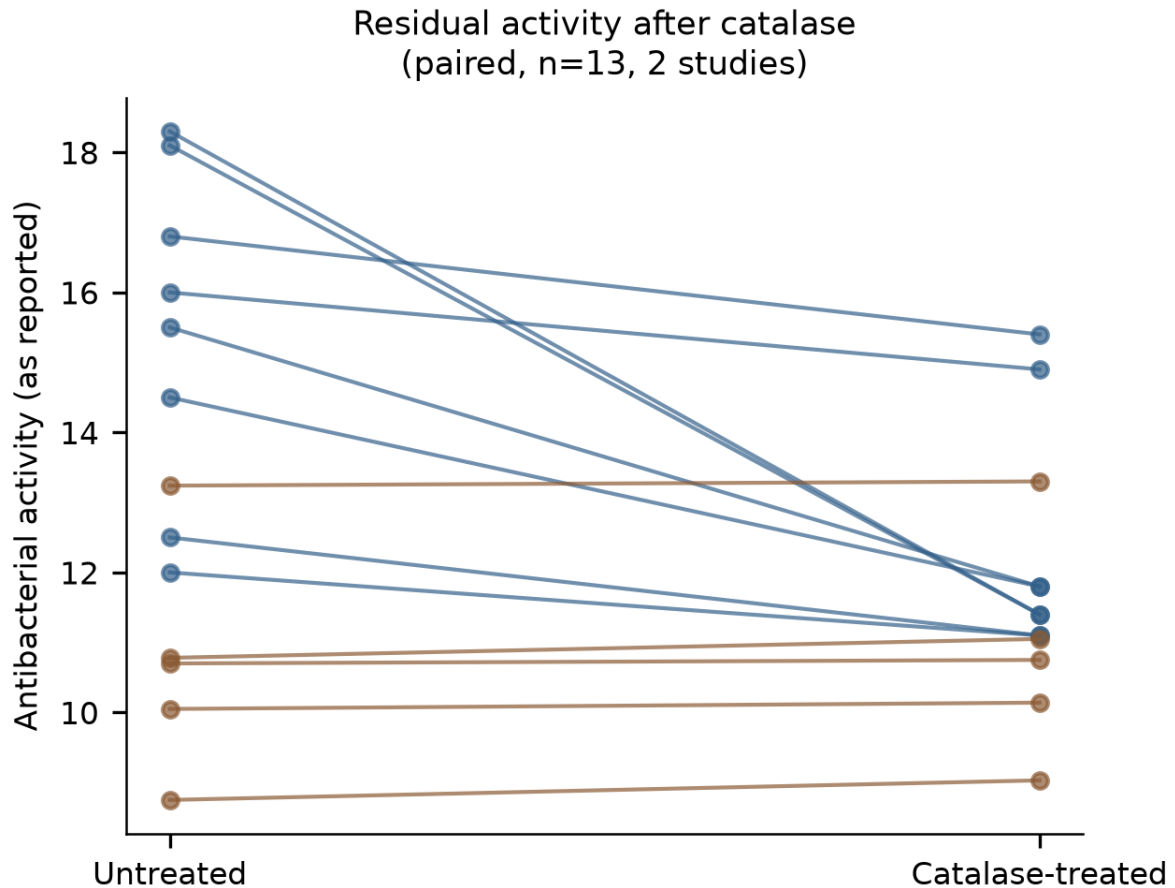


Fig. 5. Paired antibacterial activity before and after catalase treatment (13 pairs, 2 studies (Hussain et al., 2019; Mohd Hilmi et al., 2020)); residual activity indicates peroxide-independent components.

3.6. Physicochemistry versus label as predictors

Honey-level physicochemical measurements could be joined to antibacterial outcomes for 685 observations from 18 studies. Within the largest single-study block, MGO content correlated inversely with MIC ($\rho = -0.52$, $n = 116$, one study), consistent with MGO-associated potency. Honey moisture correlated positively with MIC ($\rho = 0.17$, $n = 112$, 3 studies), the expected direction if dilute, high-moisture honeys are less active. Honey pH showed no monotone association with MIC ($\rho = 0.07$, $n = 359$) but a moderate inverse association with zone size ($\rho = -0.52$, $n = 53$).

A linear model on $\log_{10}(\text{MIC})$ fitted to the largest complete-case block (116 observations, predictors MGO, pH, TPC) explained 30% of variance in-sample, versus 0% for a manuka/medical-grade label alone. However, this in-sample fit rested almost entirely on 1 study. Grouped validation told a different story: under leave-one-study-out cross-validation ($n = 155$, 3 studies, predictors pH, TPC), chemistry-based predictions performed far worse than a null model

(cross-validated $R^2 = -32.32$ vs. -0.12), and the label model was indistinguishable from null. Measured chemistry is therefore informative within individual studies but the fitted relationships did not transfer across studies in this dataset — a result we report explicitly because it bounds what cross-study chemistry-outcome synthesis can currently claim.

3.7. Organism heterogeneity

MIC distributions differed strongly across organisms (Kruskal-Wallis $H = 170$, $p < 0.001$; Fig. 6). Gram-positive organisms tended to be inhibited at lower honey concentrations (median MIC: *S. epidermidis* 11.25%, *S. aureus* 12.5%, MRSA 12.5%) than Gram-negative organisms (*E. coli* 20.0%, *P. aeruginosa* 21.0%). Manuka-type honeys showed a lower median MIC than other honeys (15% vs. 20%; Mann-Whitney $p = 0.011$). Leave-one-study-out resampling returned non-significant comparisons in 38% of iterations, so the manuka-label contrast is directionally consistent but remains sensitive to individual studies (Table 1).

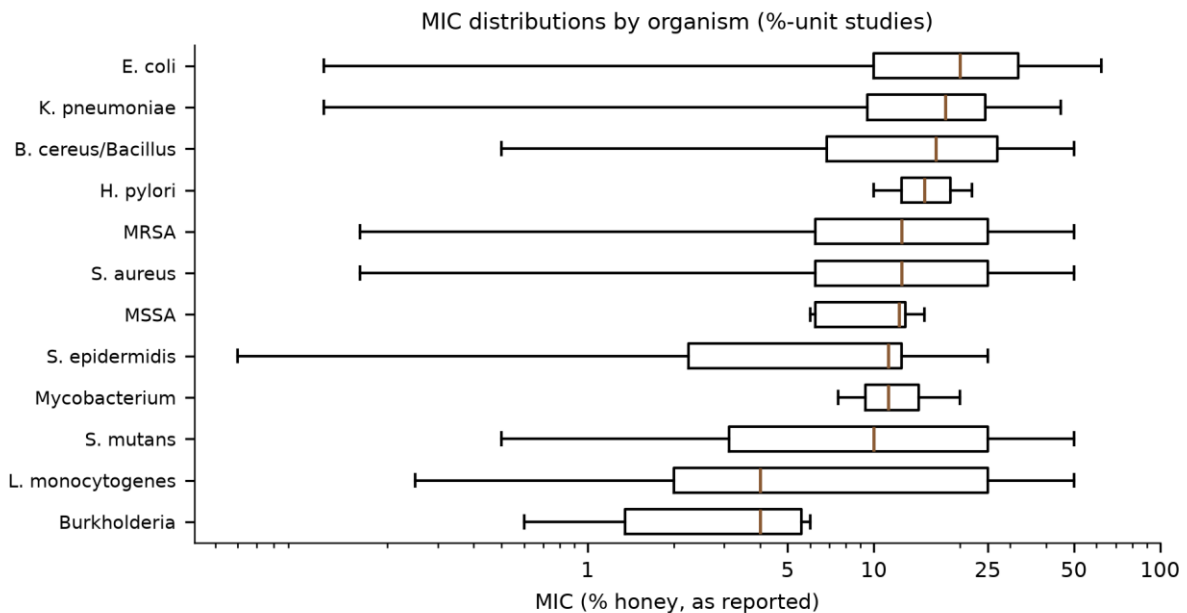


Fig. 6. MIC distributions by organism group (percentage-unit studies, log scale).

Table 1. Median MIC (% honey) by organism group, percentage-unit studies.

Organism group	Median MIC (%)
Burkholderia	4.0
L. monocytogenes	4.0
S. mutans	10.0
Mycobacterium	11.25
S. epidermidis	11.25
MSSA	12.25
S. aureus	12.5
MRSA	12.5

H. pylori	15.0
B. cereus/Bacillus	16.5
K. pneumoniae	17.75
E. coli	20.0
other	20.0
Enterobacter	21.0
P. aeruginosa	21.0

Note: MIC, minimum inhibitory concentration; MRSA/MSSA, methicillin-resistant/susceptible *S. aureus*. Values are medians of extracted MIC rows restricted to percentage-unit bases.

3.8. Sensitivity analyses

Restricting MIC analyses to the low-risk-of-bias subset left the qualitative pattern unchanged (6 studies, 185 observations, median MIC 25%). Analyzed separately by reported concentration basis, median MIC was 15% on the percent weight/volume basis ($n = 315$) and 8% on the percent volume/volume basis ($n = 157$), confirming that bases are not interchangeable and must not be pooled. Excluding manuka-labelled honeys shifted the overall median MIC only slightly (20.0%; $n = 1932$). Restricting to studies that directly measured water activity (2 studies) or H₂O₂ (7 studies) gave medians of 25% and 16% respectively. No extracted rows came from preprint records, so a peer-reviewed-only analysis is identical to the primary set. The catalase and artificial-honey analyses could not be broadened beyond the studies that performed those experiments. Table 2 summarizes the paired mechanistic contrasts.

Table 2. Paired evidence for mechanistic contrasts.

Contrast	n pairs/obs	n studies	Effect	p value
Natural vs artificial honey (potency ratio)	21	6	16/21 pairs > 1; median 1.38	0.005
Natural vs artificial honey, zone difference (mm)	10	2	median +0.6 mm	0.002
Catalase residual activity	13	2	median 92% retained	-
Activity vs concentration (series)	10	3	10/10 positive; median rho = 1.00	-
MGO vs MIC (within-study)	116	1	rho = -0.52	< 0.001
pH vs inhibition zone	53	2	rho = -0.52	< 0.001

Note: Potency ratio > 1 indicates greater natural-honey activity (zone: natural/artificial; MIC: artificial/natural). MGO, methylglyoxal; MIC, minimum inhibitory concentration.

4. Discussion

This synthesis is consistent with a dilution-dependent, multi-mechanism account of honey's antibacterial action. At high concentration, activity is compatible with dominance of water-activity/osmotic and acidity constraints: artificial sugar-equivalent honey recapitulated much — though not all — of natural honey's effect, solution pH rose monotonically toward neutrality on dilution in the one complete ladder available, and higher-moisture honeys tended to have higher MICs. On dilution those constraints weaken, and glucose oxidase can generate H₂O₂ under aqueous conditions; the limited within-study evidence located peroxide activity preferentially at low concentrations (Bang et al., 2003). Finally, catalase-resistant residual activity (median ~92% in two studies) and the within-study MGO-MIC association are compatible with a peroxide-independent component in at least some honeys. The three mechanism classes are therefore plausibly reweighted — not merely attenuated — along the dilution axis.

For food science applications the practical implication is that assay concentration is mechanistically informative, not only quantitative. An MIC measured near the top of the concentration range may reflect constraints that disappear on dilution, whereas activity at intermediate concentrations may be carried by different chemistry. Evaluations of honey as an antimicrobial ingredient or preservative should therefore report the concentration range tested and, where possible, pair outcome measurement with the candidate mechanisms (pH, water activity, H₂O₂, MGO) across the same dilution ladder.

Several findings bound what can be claimed. Cross-study chemistry-outcome correlations are observational and vulnerable to between-study confounding — the positive H₂O₂-MIC correlation is the clearest example. More importantly, grouped cross-validation showed that the apparent predictive value of measured chemistry did not survive leaving studies out: within-study gradients (notably the single-study MGO-MIC association) do not transfer across studies, and honey-type labels performed no better than a null model. Any claim that measured chemistry predicts potency across the literature would therefore overstate the evidence; what the data support is within-study association plus explicit uncertainty about generalization.

Other limitations qualify these results. First, quantitative extraction depended on open-access full texts and machine-readable tables: 103 of 412 eligible studies were extracted, and the extracted subset skews recent and open-access. Second, concentration-series data were rare, catalase pairs came from two studies, and no extracted dataset reported H₂O₂ across a dilution ladder with matched antibacterial endpoints; the peroxide-dilution hypothesis is supported qualitatively only. Third, assay heterogeneity (broth dilution, agar diffusion, differing inocula and media) limits pooling, and we deliberately avoided meta-analytic pooling across incompatible measurement bases. Fourth, automated table extraction, although validated against manual curation, may retain residual errors; all rows carry provenance for audit. Fifth, the risk-of-bias framework is a reporting-quality proxy, not a validated instrument, and the review was not prospectively registered.

Within these limits, the evidence supports a model in which concentrated honey relies substantially on physicochemical constraints, partial dilution enables peroxide-associated activity, and non-peroxide components sustain residual activity in some honeys. It does not demonstrate a

universal mechanistic switch, and the decisive experiment — matched dilution ladders of H₂O₂, water activity, pH and antibacterial endpoints within single studies — remains to be performed at scale.

5. Conclusions

Honey's antibacterial activity increased consistently with concentration in every extractable series, while the physicochemical constraints plausibly responsible for high-concentration activity weakened on dilution. Natural honey exceeded sugar-equivalent controls in most matched pairs, catalase-resistant residual activity was substantial where measured, and within-study chemistry-outcome associations — though real — did not generalize across studies. The evidence is consistent with dilution-dependent reweighting of osmotic/acidic, peroxide and non-peroxide mechanisms, but it does not establish a universal switch. Future studies should report full dilution ladders with matched physicochemical and antibacterial measurements, including concentration basis, so that mechanistic claims can be tested within studies rather than inferred across them.

CRedit authorship contribution statement

[Author One]: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing — original draft. [Author Two]: Validation, Investigation, Writing — review & editing. [Corresponding Author]: Supervision, Project administration, Writing — review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All extracted data, the data dictionary, screening decisions, analysis code and figure scripts are released with this submission as supplementary material and repository archive. Raw database search responses are included; no primary experimental data were generated for this study.

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Declaration of generative AI use

During the preparation of this work the authors used a large language-model coding assistant to develop the reproducible search, extraction and analysis code and to assist with drafting. The authors reviewed and edited all content and take full responsibility for the integrity of the publication.

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The complete list of the 103 studies included in the quantitative synthesis is given in Supplementary Table S11.